MODULE 2

LADDER PROGRAMMING GUIDE

Siemens TIA Portal PLC Course
Module 2: Introduction to Siemens PLC

Author: AUTOMATION AND ROBOTICS ENGINEER
Level: Basic
Software: Siemens TIA Portal
Language: Ladder Diagram (LAD)

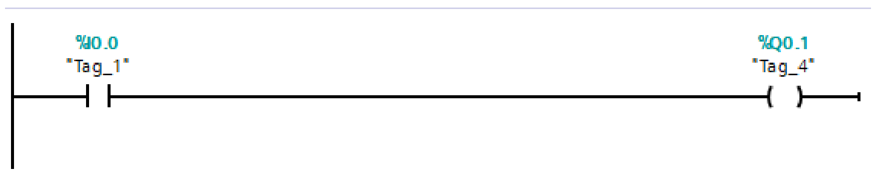
Industrial Automation and Control

This technical guide presents the essential fundamentals of Ladder programming used in modern industrial automation systems. The content is intended for technical students, university students, and professionals in the field of Industrial Automation and Control. It explains the main elements of the Ladder language used in Siemens PLCs and other industrial platforms.

IMAGES ARE TAKEN FROM TIA PORTAL V20.

1. Normally Open Contacts (NO)

Normally open contacts allow logical continuity when the associated variable is energized. In Ladder programming they represent sensors, push buttons, or active logic conditions.



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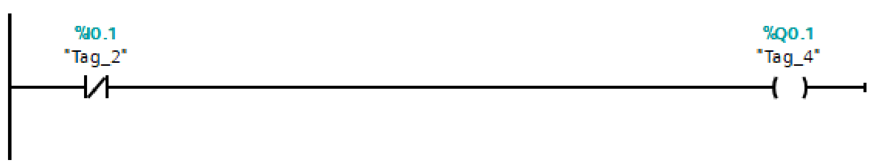
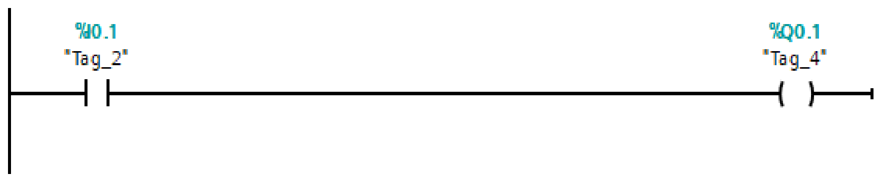
2. Normally Closed Contacts (NC)

Normally closed contacts allow logical continuity while the variable is de-energized. They are widely used in safety circuits, emergency stops, and fault conditions.



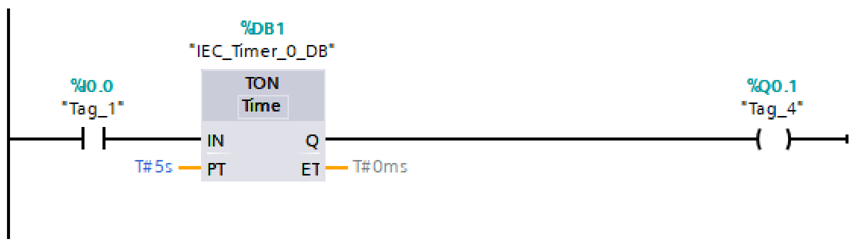
3. Coils

Coils represent digital outputs or internal memory bits within the PLC. When the logic of the rung is true, the coil energizes.



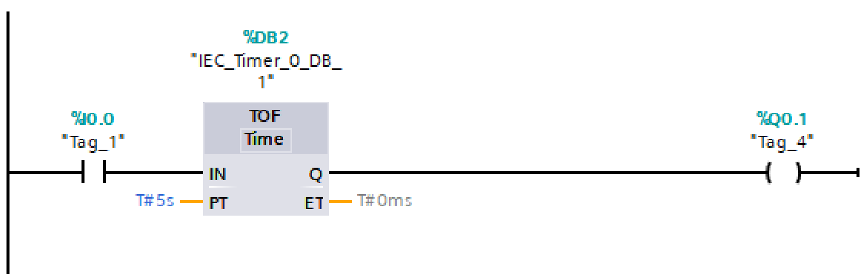
4. TON Timer (On Delay)

The TON timer activates its output after a programmed time has elapsed since the input was energized. It is used for delayed starts, automatic sequences, and process control.



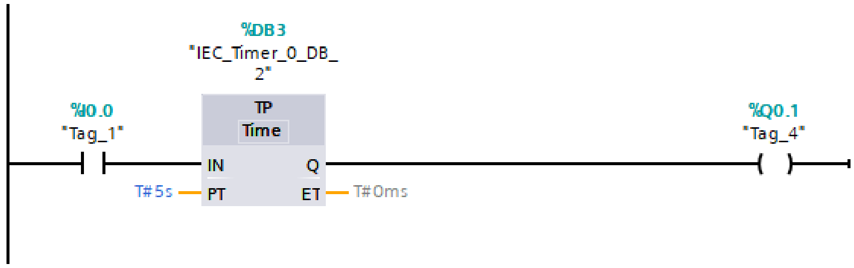
5. TOF Timer (Off Delay)

The TOF timer keeps the output activated for a set time after the input signal is deactivated. It is used for residual ventilation, cooling, and post-shutdown timing.



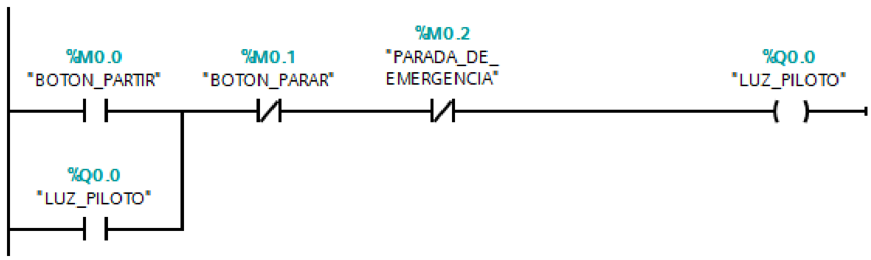
6. TP Timer (Pulse Timer)

The TP timer generates a fixed-duration pulse when it detects a triggering edge. It is used for pulse generation, momentary activation, and automatic sequences.



7. Industrial Latching (Seal-in Circuit)

Latching is a technique widely used to keep an output activated even after the start button is released. It is used in motors, pumps, conveyor belts, and automatic systems.



Technical Conclusion

Ladder programming remains one of the most widely used languages in industrial automation due to its ease of interpretation and maintenance. Mastering contacts, coils, timers, and latching circuits forms a fundamental basis for developing reliable and safe automated systems.